IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2024

ANALYSIS AND DESIGN OF CIRCUITS

Thursday, 2 May 2:30 pm

Time allowed: 2:00 hours

There are THREE questions on this paper.

Answer ALL questions. Question One carries 40% of the marks. Question 2 and 3 carry equal marks (30% each)

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : Z. Akhtar, P. Mitcheson
Second Marker(s) :

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Information for Candidates

Numerical answers

- Numerical answers should be given as fully evaluated decimal values and not as unevaluated arithmetic expressions.

Notation

The following notation is used in this paper:

- The voltage waveform at node X in an AC circuit is denoted by $x(t)$ while the phasor voltage is denoted by X.

- Component and source values are generally given in Ohms, Farads, Henrys, Volts or Amps with the unit symbol omitted. Where an imaginary number is specified, it represents the complex impedance or phasor value.

- Times are in seconds unless otherwise stated.

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1. a) Using nodal analysis, calculate the voltages at nodes *X* and *Y* of *Fig. 1a*.

[6]

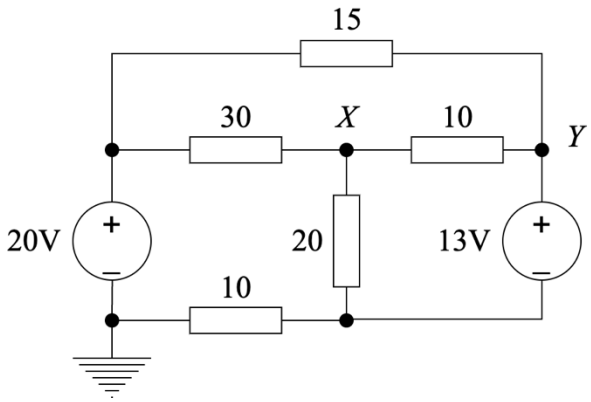


Fig. 1a

b) Using the **principle of superposition**, find the Thévenin's equivalent of the circuit between nodes A and B in *Fig. 1b*.

[6]

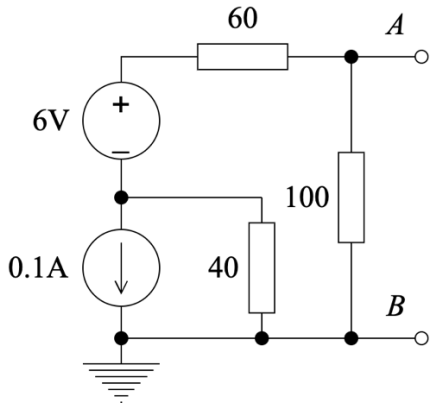


Fig. 1b

c) Calculate the complex power associated with the resistor, inductor and source in *Fig. 1c* when $C = 0$. Then, determine the value for the capacitor C that will render the complex power delivered by the source purely real. The source has a frequency of 50 Hz.

[8]

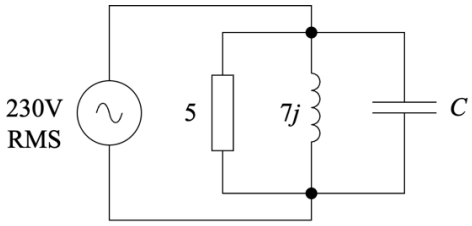


Fig. 1c

d) For the amplifier of *Fig. 1d*, calculate values of R_B and R_C that set the quiescent value of V_{out} at 5 V with 1.5 mA of collector current. Draw a small signal model and calculate the small signal voltage gain. Do not ignore R_B in your calculations.

[8]

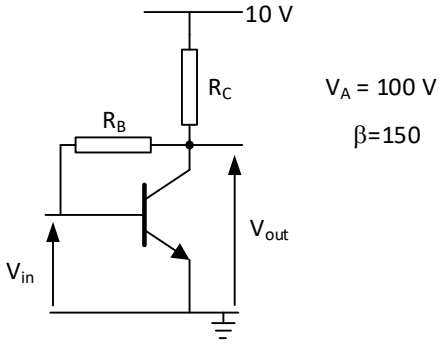


Fig. 1d

e) Draw a class AB amplifier arrangement, explain its use and state why it is superior to a class B stage. You may wish to draw a sketch graph of the input and output voltage for both class B and class AB stages to explain your answer.

[5]

f) A new type of transistor called the Imperial-FET, shown in *Fig. 1f*, has been proposed. It has a circuit symbol shown below. The terminals are called *control* (c), attractor (a) and *repeller* (r). current is considered positive into the attractor and out of the *repeller*. There is no current flow into the control terminal.

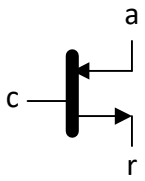


Fig. 1f

The current into the attractor, I_a, is controlled solely by the Control-Repeller Voltage, V_{cr}, by the equation:

$$I_a = \sin\left(\frac{\pi}{2}V_{cr}\right)$$

Taking inspiration from the small signal MOSFET model, draw a small signal model of the Imperial-FET, and derive any small signal parameters, explaining your method.

[7]

2. a) The circuit shown in *Fig. 2a* is a filter.

i) Write down the transfer function $\frac{Y}{X}(j\omega)$ for the circuit in *Fig. 2a*.

[5]

ii) Find the high and low frequency asymptotes of the transfer function to show that the circuit is a high pass filter.

[3]

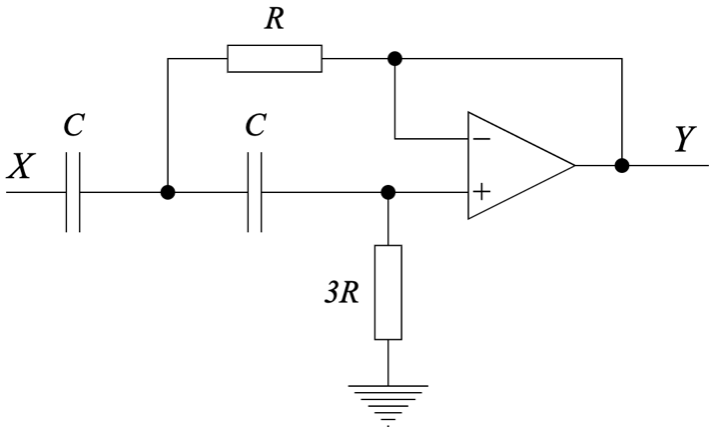


Fig. 2a

b) For the circuit shown in *Fig. 2b*, write down the transfer function $H = \frac{Y}{X}(j\omega)$ and sketch a straight-line approximation to the frequency plot of $|H|$ between 10 Hz and 100 kHz. [Only the magnitude plot of $|H|$ is required. You do not need to plot the phase.]

[10]

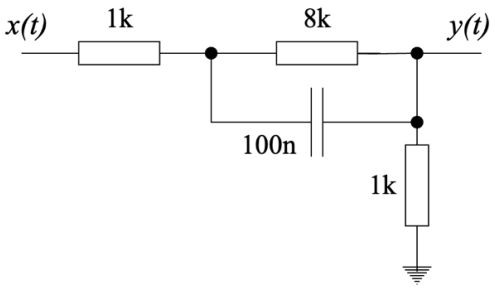


Fig. 2b

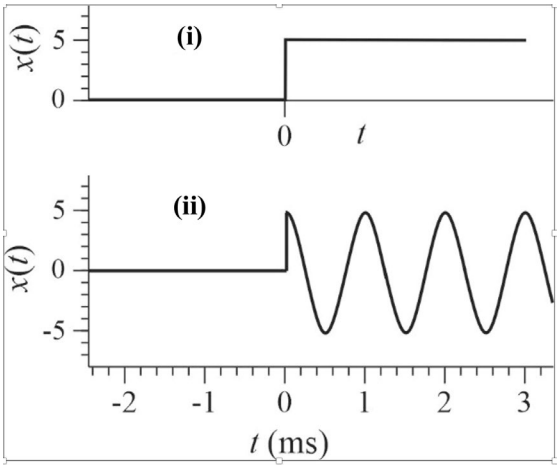


Fig. 2c

c) For the circuit shown in *Fig. 2b*, obtain an expression for the time variation of the output voltage $y(t)$ when the input signal $x(t)$ is as shown in *Fig. 2c (i)*, with $x(t) = 0$ for $t < 0$ and $x(t) = 5$ for $t \geq 0$.The initial charge on the capacitor is zero. Draw a dimensioned sketch of $y(t)$ covering the time interval $-0.5\ \text{ms} \leq t \leq 2\ \text{ms}$.

[6]

d) For the circuit shown in *Fig. 2b*, using the transfer function method, obtain an expression for the time variation of $y(t)$ when the input signal $x(t)$ is as shown in *Fig. 2c (ii)*, with $x(t) = 0$ for $t < 0$ and $x(t) = 5\cos(2000\pi t)$ for $t \geq 0$.

[6]

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3.

a) Derive a formula for the small signal voltage gain of the common emitter amplifier arrangement of *Fig. 3a* in terms of V_{CC} and V_T, when the quiescent output voltage is biased at $V_{cc}/2$. The gain is independent of the value of R_C. Why, and what are the implications of this? You may assume the transistor's output resistance is infinite.

[5]

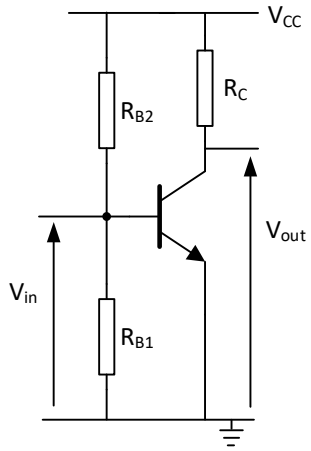


Fig. 3a

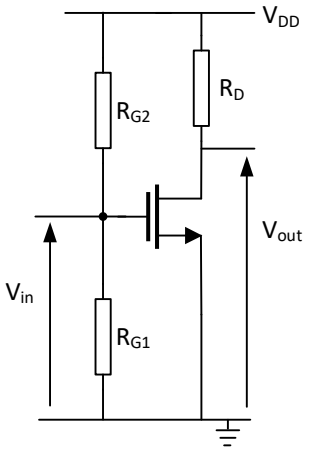


Fig. 3b

b) Derive a formula for the small signal voltage gain of the common drain amplifier arrangement of *Fig. 3b*, assuming the quiescent output voltage is biased at $V_{DD}/2$. Assume the conductivity parameter of the MOSFET is k_1. Does the magnitude of the small signal voltage gain increase, or decrease with R_D? Assume the transistor output resistance is infinite.

[5]

e) *Fig. 3c* shows a MOSFET based differential amplifier.

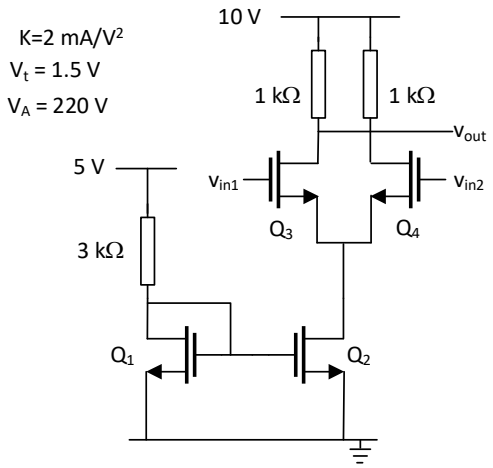


Fig. 3c

i) Draw a small signal model for the circuit in differential mode excitation and calculate the differential voltage gain to the single ended output, V_{out}, when the input voltage is taken as (V_{in1}-V_{in2}). Ensure you have the correct sign.

[8]

ii) Calculate the common mode voltage gain of the amplifier and the common mode rejection ratio. You may assume that under common-mode excitation, the change in V_{GS} for transistors Q_3 and Q_4 is negligible.

[4]

iii) Comment on the performance of this differential gain stage. Is it useful?

[2]

f) It is proposed to increase the gain of the differential amplifier by connecting the output of *Fig. 3c* to the input of *Fig. 3a* using capacitive coupling.

i) Calculate the input and output impedance of *Fig. 3c* and draw a macro model for the amplifier in differential mode.

[3]

ii) Assume that *Fig. 3a* has been designed with a voltage gain of -100, an input impedance of 100 kΩ and an output impedance of 1 kΩ. Draw a macro model of the system where the output of *Fig. 3c* is capacitively coupled to the input of *Fig. 3a*, which in turn drives a 2 kΩ load. Calculate the mid-band gain.

[3]